

Rule-Based Labeling of Risky Driving Behavior

In our driving datasets, labels indicating *risky* or *normal* driving behavior were unavailable. To address this limitation, we adopted a **rule-based labeling strategy** grounded in **domain knowledge** and **data-driven statistical analysis**.

The objective of this step is to:

- Define what constitutes *risky driving* in a **quantitative and reproducible way**
 - Generate reliable labels to enable **supervised machine learning**
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2. Statistical Analysis of Driving Features

2.1 Descriptive statistics

We first analyzed the statistical distribution of each behavior-related feature using descriptive statistics, including:

- minimum and maximum values
- mean and standard deviation
- percentile values (25%, 50%, 75%, 90%, 95%, 99%)

This was achieved using:

```
df_features.describe(percentiles=[0.25, 0.5, 0.75, 0.9, 0.95, 0.99])
```

Purpose of percentiles

Percentiles allow us to identify **extreme values** in the data:

- Upper percentiles (90%–99%) represent **unusually high intensity events**

- Lower percentiles (1%–10%) represent **unusually low or extreme negative events**

Such extremes are commonly associated with unsafe driving behavior.

3. Threshold Definition Strategy

3.1 Principle used

Rather than using arbitrary fixed values, **thresholds were derived directly from the dataset** using percentile-based rules.

This ensures that:

- Thresholds adapt to the specific driving population
 - Labeling remains objective and data-driven
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3.2 Feature-wise threshold definition

Feature	Risky behavior	Threshold rule	Justification
Vehicle_speed	Overspeeding	95th percentile	Very high speed relative to normal driving
Acceleration_speed_-_Longitudinal	Harsh braking	5th percentile	Extreme negative acceleration
Acceleration_speed_-_Longitudinal	Hard acceleration	95th percentile	Sudden high acceleration
Acceleration_speed_-_Lateral	Aggressive turning	95th percentile	High lateral force during turns
Steering_wheel_angle	Sharp steering	95th percentile of absolute value	Sudden large steering angles
Steering_wheel_speed	Sudden steering	95th percentile of absolute value	Fast steering wheel movements

Master_cylinder_pressure Hard braking 95th percentile Strong braking force

These thresholds were computed as:

```
thresholds['Vehicle_speed'] =  
df_features['Vehicle_speed'].quantile(0.95)  
thresholds['Harsh_braking'] =  
df_features['Acceleration_speed_-_Longitudinal'].quantile(0.05)  
thresholds['Hard_acceleration'] =  
df_features['Acceleration_speed_-_Longitudinal'].quantile(0.95)  
thresholds['Aggressive_turn'] =  
df_features['Acceleration_speed_-_Lateral'].quantile(0.95)  
thresholds['Sharp_steering'] =  
df_features['Steering_wheel_angle'].abs().quantile(0.95)  
thresholds['Sudden_steering'] =  
df_features['Steering_wheel_speed'].abs().quantile(0.95)  
thresholds['Hard_braking_pressure'] =  
df_features['Master_cylinder_pressure'].quantile(0.95)
```

4. Visualization of Risk Thresholds

To validate the thresholds visually, distributions of each feature were plotted along with their corresponding threshold.

Example for vehicle speed:

```
sns.histplot(df_features['Vehicle_speed'], bins=50, kde=True)  
plt.axvline(thresholds['Vehicle_speed'], color='r', linestyle='--')
```

Interpretation

- The red dashed line represents the **risk threshold**
- Data points to the right of this line correspond to **extreme driving events**
- Similar plots were generated for all selected features

This step ensures that thresholds align with intuitive and physical interpretations of risky behavior.

5. Rule-Based Labeling Method

5.1 Binary risk label definition

Each driving record (row) was labeled as **risky** if **at least one risky behavior condition** was satisfied

Implemented as:

```
df_features['risky'] = (
    (df_features['Vehicle_speed'] > thresholds['Vehicle_speed']) |
    (df_features['Acceleration_speed_-_Longitudinal'] <
thresholds['Harsh_braking']) |
    (df_features['Acceleration_speed_-_Longitudinal'] >
thresholds['Hard_acceleration']) |
    (df_features['Acceleration_speed_-_Lateral'] >
thresholds['Aggressive_turn']) |
    (df_features['Steering_wheel_angle'].abs() >
thresholds['Sharp_steering']) |
    (df_features['Steering_wheel_speed'].abs() >
thresholds['Sudden_steering']) |
    (df_features['Master_cylinder_pressure'] >
thresholds['Hard_braking_pressure'])
).astype(int)
```

6. Feature-Level Risk Indicators

To analyze **which behaviors contribute most to risk**, separate binary indicators were created for each risky event type:

```
df_features['Risky_speed'] = ...
df_features['Harsh_braking'] = ...
df_features['Hard_acceleration'] = ...
...
```

Each column indicates whether a specific risky behavior occurred.

7. Risk Contribution Analysis

The total number of risky events per feature was computed and visualized:

```
risky_summary = df_features[...].sum()
```

Interpretation

- Features with higher counts contribute more frequently to risky behavior
- This provides insight into **dominant risk factors** in the dataset
- The bar plot offers a clear, interpretable summary of behavioral risk patterns